MAKSIM ZHDANOV

email · website · github · google scholar · twitter · linkedin

research: sparse attention; long-context modeling; hardware-aligned architectures; large-scale weather forecasting

EDUCATION

PhD in Machine Learning 2023 – 2027
University of Amsterdam, AMLab
· Advisors: Max Welling & Jan-Willem van de Meent

MSc in Computer Science 2019 – 2022
TU Dresden (GPA 1.4, excellent)

BSc in Physics 2015 – 2019
Saint Petersburg State University (GPA 4.8/5.0, with honours)

EXPERIENCE

Research Intern Jun 2026 – present
Google DeepMind, London
· Weather team, hosted by Ferran Alet & Ilan Price
· Hardware-aligned architectures for large-scale weather forecasting

Research Intern Feb 2026 – May 2026
CuspAI, Amsterdam
· Hosted by Govert Verkes
· Agentic search for materials discovery

Research Assistant Sep 2020 – Apr 2023
Helmholtz AI, Dresden
· ML for neuroscience and materials science

PUBLICATIONS

Native Multi-Dimensional Subquadratic Operators via Input Dependent Long Convolutions
David R. Wessels, Farhad Ramezanghorbani, Alireza Moradzadeh, David W. Romero, Olivia Viessmann, Maksim Zhdanov, John St. John, Ken Janik, David M. Knigge, Yucheng Tang, Erik J. Bekkers, Saee Gopal Paliwal
NeurIPS 2026 arxiv

Follow the Mean: Reference-Guided Flow Matching
Pedro M. P. Curvo, Maksim Zhdanov, Floor Eijkelboom, Jan-Willem van de Meent
NeurIPS 2026 arxiv

(Sparse) Attention to the Details: Preserving Spectral Fidelity in ML-based Weather Forecasting Models
Maksim Zhdanov, Ana Lucic, Max Welling, Jan-Willem van de Meent
ICML 2026; Best Paper Runner-up, FM4S @ ICLR 2026 arxiv weights demo

Conditional Clifford-Steerable CNNs with Complete Kernel Basis for PDE Modeling
Bálint László Szarvas*, Maksim Zhdanov*
ICML 2026 arxiv code

MSPT: Efficient Large-Scale Physical Modeling via Parallelized Multi-Scale Attention
Pedro Curvo, Jan-Willem van de Meent, Maksim Zhdanov
CVPR 2026 (Highlight) arxiv code

AdS-GNN: a Conformally Equivariant Graph Neural Network
Maksim Zhdanov, Nabil Iqbal, Erik Bekkers, Patrick Forré
ICLR 2026 arxiv code

Adaptive Mesh Quantization for Neural PDE Solvers
Winfried van den Dool*, Maksim Zhdanov*, Yuki Asano, Max Welling
TMLR 2025 arxiv code

Erwin: A Tree-based Hierarchical Transformer for Large-scale Physical Systems
Maksim Zhdanov, Max Welling, Jan-Willem van de Meent
ICML 2025 arxiv code blog

Clifford-Steerable Convolutional Neural Networks
Maksim Zhdanov, David Ruhe, Maurice Weiler, Ana Lucic, Johannes Brandstetter, Patrick Forré
ICML 2024 arxiv code blog

Implicit Convolutional Kernels for Steerable CNNs
Maksim Zhdanov, Nico Hoffmann, Gabriele Cesa
NeurIPS 2023 arxiv code blog

Investigating Brain Connectivity with Graph Neural Networks and GNNExplainer
Maksim Zhdanov, Saskia Steinmann, Nico Hoffmann
ICPR 2022 (Oral) arxiv code

* equal contribution

WORKSHOP PAPERS

BSA: Ball Sparse Attention for Large-scale Geometries
Catalin E. Brita, Hieu Nguyen, Lohithsai Yadala Chanchu, Domonkos Nagy, Maksim Zhdanov
LCFM @ ICML 2025 arxiv code

Amortized Bayesian Inference of GISAXS Data with Normalizing Flows
Maksim Zhdanov, Lisa Randolph, Thomas Kluge, Motoaki Nakatsutsumi, Christian Gutt, Marina Ganeva, Nico Hoffmann
ML4PS @ NeurIPS 2022 arxiv

Learning Generative Factors of EEG Data with Variational Auto-Encoders
Maksim Zhdanov, Saskia Steinmann, Nico Hoffmann
DGM4MICCAI 2022 arxiv

OPEN-SOURCE CONTRIBUTIONS

UMA Nitro 2026
· Contributed to a faster inference implementation of UMA in Meta FAIR's fairchem

INVITED TALKS

Triton & Flash Attention Workshop, ELLIS Deep Thinking Hour Dec 2025
AI4Science Reading Group, Mila, Montréal Aug 2025

TEACHING

Teaching Assistant, University of Amsterdam 2023 – 2025
· Machine Learning I (2023); Deep Learning II (2024, 2025)
· Guest lecture: Introduction to Triton, kernel fusion and Flash Attention (2026)

TECHNICAL SKILLS

Code: Python, C++, Triton **ML:** JAX, PyTorch; large-scale TPU/GPU training